

In this Standard the comma has been adopted as the decimal sign.

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NE 03-071

MACHINED METALLIC DETAILS – GENERAL TOLERANCES

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ISSUE DATE: 20 Dec 23

REVISION REPORT: RR_NE 03-071L

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1. SCOPE

This Standard establishes general tolerances (dimensional and geometric) for machined details of aeronautical parts.

2. APPLICATION

This Standard shall be applied, whenever it is called out on drawing (see item 7.1), for dimensions or geometric configurations without specific tolerances or if there are not tolerances according to other mandatory standards.

NOTE

If the tolerances established in this Standard do not guarantee the function or interchangeability of the part, specific tolerances shall be indicated on drawing.

3. APPLICABLE DOCUMENTS

Embraer Standards

NE 03-004 - Surface Roughness.

NE 03-006 - Riveting - Drilling, Countersinking and Dimpling.

NE 03-075 – System of Limits and Fits – ISO 286.

NE 03-085 – Surface Roughness – Inspection.

4. DEFINITIONS

4.1 GENERAL TOLERANCE

Common tolerance to several dimensions or geometric configurations. It is not indicated next to the nominal dimension on the drawing.

4.2 SPECIFIC TOLERANCE

Tolerance indicated next to the nominal dimension or configuration on drawing.

4.3 CORNER INTERNAL RADIUS

Machined internal radius, obtained by milling cutter or by its radial displacement. See Figure 1.

4.4 CORNER EXTERNAL RADIUS

External radius between two surfaces. See Figure 1.

4.5 SHARP INTERNAL CORNER

Intersection, without chamfer or radius between two machined internal surfaces. See Figure 1.

4.6 SHARP EXTERNAL CORNER

Intersection, without chamfer or radius, between two machined surfaces (at least one of them shall be external surface). See Figure 1.

4.7 BROKEN EDGE

Small chamfer or radius along the corner. See Figure 1.

4.8 EXTERNAL CHAMFER

Dimensioned machined detail in order to eliminate a sharp edge. See Figure 1.

4.9 BOTTOM INTERNAL RADIUS

Internal radius machined by the corner radius of milling cutter. See Figure 1.

4.10 MISMATCH

Machining deviation on a same surface or on an intersection between two surfaces. It is caused by tool position changes or by the use of two tools. See Figure 1.

4.11 PERIPHERICAL TEXTURE (SURFACE ROUGHNESS)

Micro geometrical deformations (salience and re-entrances) of a machined surface. They are caused by cutting tool action mainly. See Figure 1.

4.12 WAVING

Medium geometrical deformations (salience and re-entrances) of a machined surface. They are caused by:

- a) too small corner radius tool;
- b) too great cutting feed;
- c) inadequate cutting speed.

See Figure 1.

4.13 HEMSTITCHING

Medium geometrical scallops (waving) arranged on lines (saliencies and re-entrances) on a machined surface by milling operation (generally). The hemstitching is caused by ball nose end mill on plain, contoured or inclined surfaces or by concordance radius of ball nose end mill on contoured or inclined surfaces. See Figure 1.

4.14 WARPING

Concomitant geometrical deformations (incurvation, torsion) of two or more surfaces of a machined part. The deformation of only one surface is a machining defect but it is not a warping.

Examples of warping causes:

- a) heat treatments before or after machining;
- b) inadequate jigs for parts fixation;
- c) inadequate cutting tools;
- d) too great cutting depth;
- e) too thin machined detail;
- f) internal residual strain in material of part.

See Figure 1.

4.15 FLATNESS

Deviation of a flat surface. See Figure 1.

4.16 BURRS

Chip residues adhered on sharp edges or drilled hole surfaces. See Figure 1.

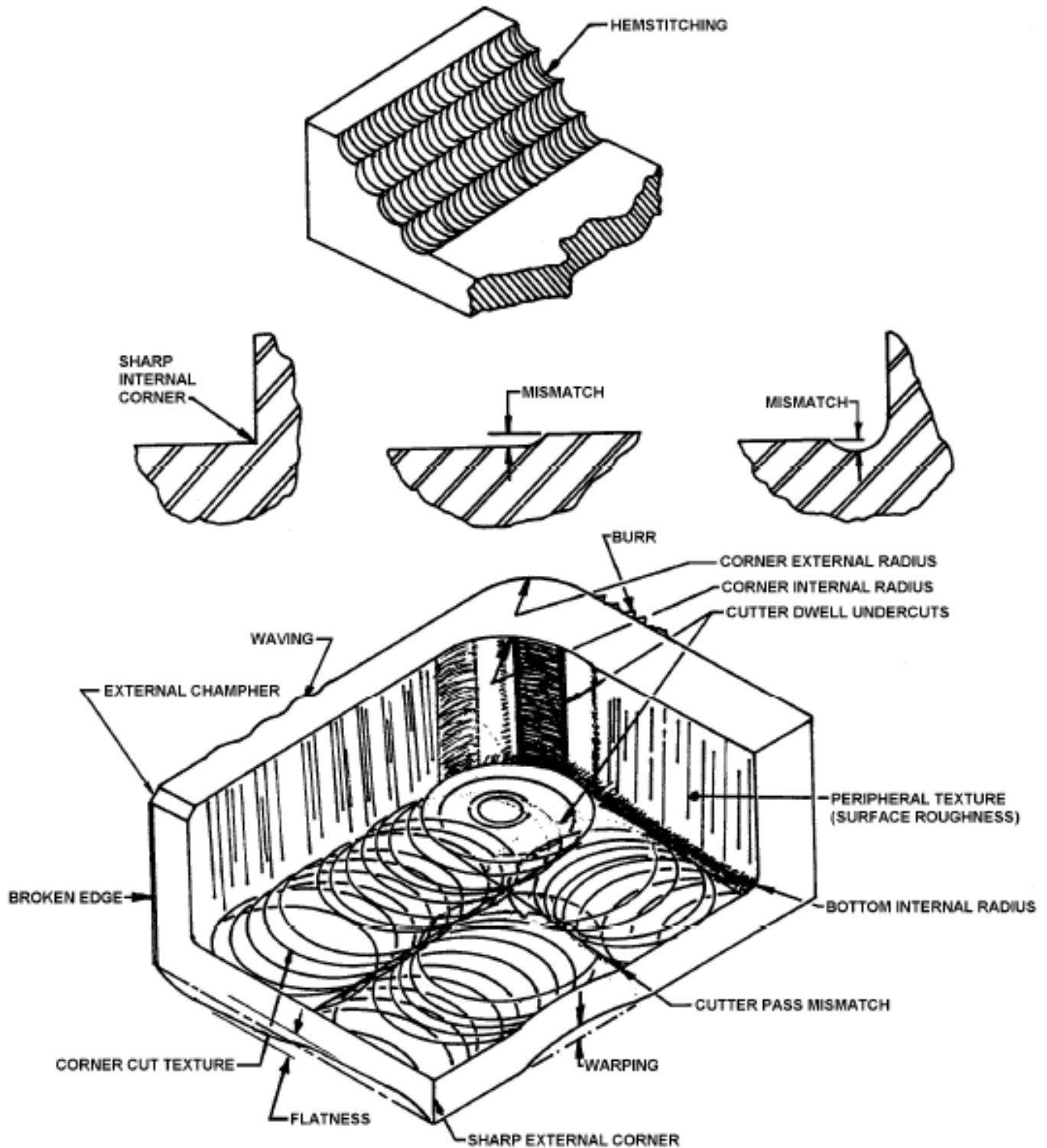


FIGURE 1 : MACHINING DETAILS AND IMPERFECTIONS

4.17 MODEL BASED DEFINITION (MBD)

Illustrative 3D drawing that captures every detail of product, project information and engineering requirements.

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5. REQUIREMENTS

5.1 GENERAL DIMENSIONS

5.1.1 Tolerances for linear or angular dimensions shall be in agreement with Table 1.

TABLE 1: GENERAL TOLERANCES FOR LINEAR DIMENSIONS

NOMINAL LINEAR DIMENSION "D" EXCEPT ROUND DIAMETER Mm (1) (2) (3) (4)	TOLERANCE mm	NOMINAL ANGULAR DIMENSION "θ" INCLUDED NON- DIMENSIONED 90°	TOLERANCE (5)
- < D ≤ 1,5	+ 0,3 - 0,1	0° < θ < 360°	± 1°
1,5 < D ≤ 3	± 0,3		
3 < D ≤ 6	± 0,4		
6 < D ≤ 30	± 0,5		
30 < D ≤ 100	± 0,8		
100 < D ≤ 2000	± 1,0		
2000 < D ≤.....	± 0,0006D		

(1) For distances between cutouts on instrument panels refer to Table 2.

(2) For radii chamfer heights refer to 5.2.

(3) For round holes refer to 5.3.

(4) Holes for fasteners installation:

- the pitch tolerance for bolt installation holes shall be according to the specific tolerance indicated on drawing;
- the distance from edge tolerance for bolt installation holes shall be according to Table 1;
- the pitch and edge distances tolerance for rivet installation holes shall be according to NE 03-006 and according specified on drawing.

(5) Use angular tolerance if it is impossible to use linear tolerance.

5.1.2 For MBD methodology, dimensions not shown in the production drawing shall be measured directly in the 3D model and for not tolerated geometries, apply the general tolerances herein specified. For the manufacturing, the dimensions shall be informed by the operations outline.

5.1.3. For machined sheet parts the specific thickness tolerance presented on Table 1A shall be applied.

TABLE 1A : SPECIFIC THICKNESS TOLERANCES APPLICABLE TO MACHINED MRS SHEET (RAW MATERIAL THICKNESS LOWER THAN 6,35 mm.) PARTS

TYPE	THICKNESS (RANGE)mm	TOLERANCE
Specific	0,8 ≤ t ≤ 1,5	- 0,1 + 0,15
	1,5 < t ≤ 3,0	±0,15
	3,0 < t ≤ 6,35	±0,2

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5.2 RADII AND CHAMFERS

Tolerance for bottom, internal corner or external corner radii and for chamfers (both sides) shall be according to Table 3.

TABLE 2: GENERAL TOLERANCES FOR DISTANCES BETWEEN CUTOUTS ON INSTRUMENT PANELS

NOMINAL LINEAR DIMENSION "D" (mm)	TOLERANCE (mm)
$D \leq 1,5$	+ 0,3 - 0,1
$1,5 < D \leq 100$	$\pm 0,3$
$100 < D \leq 400$	$\pm 0,4$
$400 < D \leq 1000$	$\pm 0,6$
$1000 < D \leq 2000$	$\pm 0,8$
$2000 < D \leq -$	$\pm 0,0006D$

TABLE 3: GENERAL TOLERANCES FOR RADII AND CHAMFERS

NOMINAL DIMENSION R = radius S = chamfer side mm (1)	TOLERANCE (mm)
$0,5 < R \text{ or } S \leq 1$ (2)	$\pm 0,3$
$1 < R \text{ or } S \leq 3$	$\pm 0,5$
$3 < R \text{ or } S \leq 6$	$\pm 1,0$
$6 < R \text{ or } S \leq 30$	$\pm 2,0$
$30 < R \text{ or } S \leq 120$	$\pm 4,0$
$120 < R \text{ or } S \leq 400$	$\pm 8,0$
$400 < R \text{ or } S \leq \dots$	$\pm 0,02R \text{ or } \pm 0,02S$

(1) Sharp edges

- sharp edges specified on drawings shall be deburred only; they shall not be broken.
- sharp edges not specified on drawings shall have $R \text{ or } S \leq 0,5 \text{ mm} \times 45^\circ$ (REF), since R or S is smaller 10% of detail thickness.
- sharp edges on hole details with thickness equal or less than 2 mm shall be broken with chamfer or radius equal or less than 5% of the thickness.

(2) R or (S) $\leq 0,5 \text{ mm}$, shown without tolerance on drawing, shall be considered as R or S max.

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5.3 ROUND HOLES

5.3.1 HOLE DIAMETER

Tolerance for hole diameter shall be according to Table 4.

TABLE 4 : GENERAL TOLERANCES FOR ROUND HOLES

NOMINAL HOLE DIAMETER Ø mm (1)	TOLERANCE OF Ø mm (3)
$- < \varnothing \leq 3$	+ 0,25 0
$3 < \varnothing \leq 6$	+ 0,30 0
$6 < \varnothing \leq 10$	+ 0,36 0
$10 < \varnothing \leq 18$	+ 0,43 0
$18 < \varnothing \leq 30$	+ 0,52 0
$30 < \varnothing \leq 50$ (2)	+ 0,62 0

(1) Centering holes (guide holes) dimensioned with ϕ 2,1 mm on drawings may be drilled (preferably) with ϕ 2,5 mm since the manufacturing cost justifies the alteration.

(2) Hole diameters larger than 50 mm shall have tolerances according to H14 of NE 03-075.

(3) Hole diameter for fasteners installation:

- hole diameters for bolt/screw installation shall be according to specific tolerance indicated on drawing;
- hole diameter for rivet installation shall be according to NE03-006 and according specified on drawing.
- 2,5 mm diameter guide holes in sheet metal primary parts, which will undergo subsequent enlargement for fastener installation in assembly line, may be approved with diametric variation of + 0,5 mm / - 1,0 mm.

5.3.2 HOLE CENTER LINE

a) tolerance for angular displacement of hole center line is $\pm 1^\circ$.

b) tolerance for linear displacement of hole center line is ± 1 mm.

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5.4 GEOMETRICAL DEVIATIONS

Any geometrical deviation detected shall be identified according to 5.4.1 through 5.4.9. In case of doubt, contact Materials or Structure Engineering.

5.4.1 SURFACE ROUGHNESS

It shall be $R_a = 3,2 \mu\text{m}$ (maximum) according to NE 03-085.

5.4.2 PROFILE OF ANY SURFACE

Tolerance for profile of any surface shall be  1 mm

5.4.3 CONTOURS

The tolerance for undimensioned contours shall be $\pm 1 \text{ mm}$.

5.4.4 MACHINING DISCORDANCE (MISMATCH)

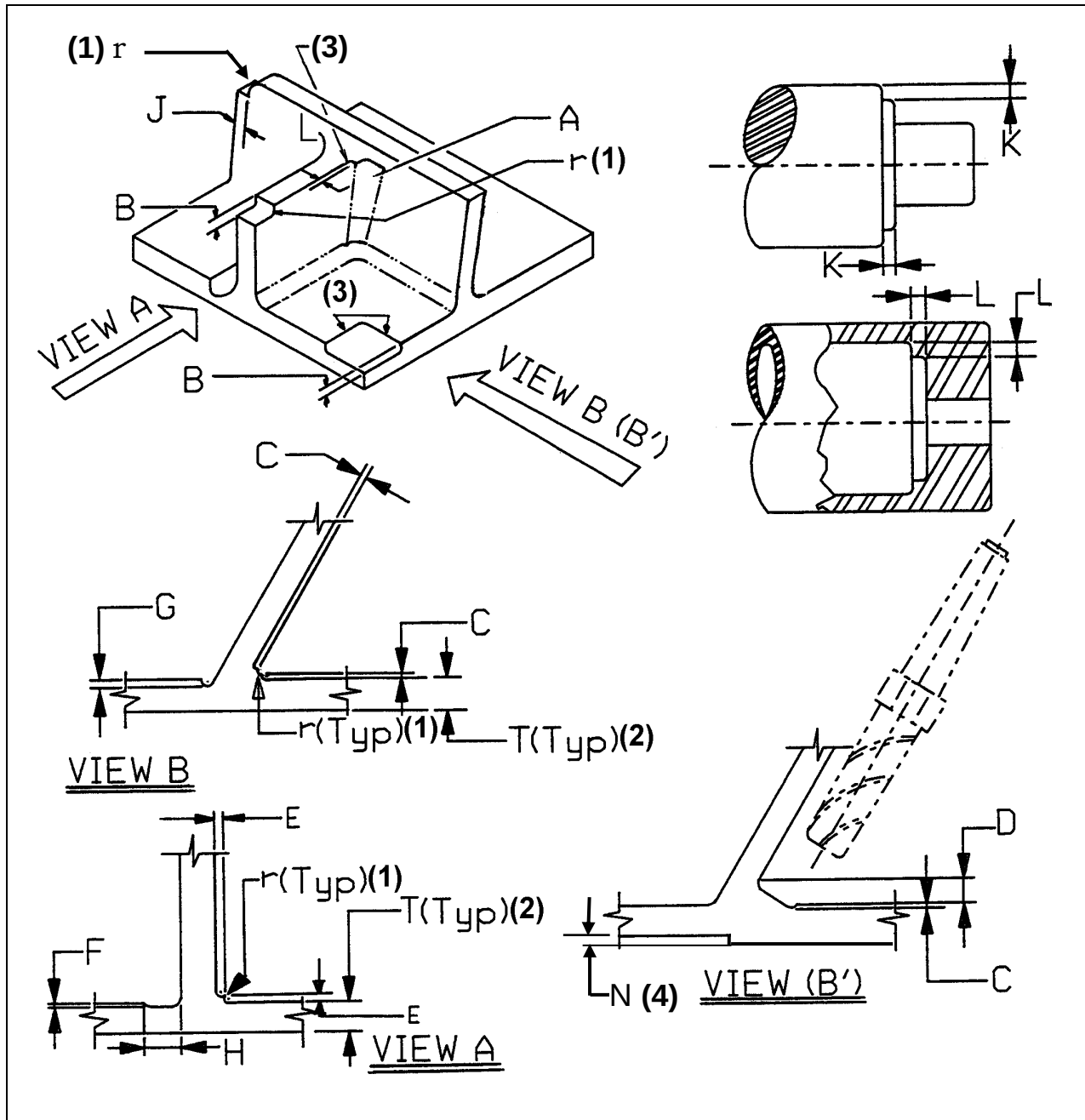
Mismatches are permitted of machined detail according to Table 5 provided;

- a) there are not limitations for them on drawing;
- b) the tolerated dimensions, on mismatch area, specified on drawing shall be achieved;
- c) the surface roughness, on mismatch area, specified on drawing shall be achieved.
- d) unless otherwise specified herein after or on drawing, the minimum acceptable radius at base of mismatch (i.e: the minimum cutter tool radius) shall be 0,8 mm. For lower radius the mismatch shall be eliminated by sanding operation.
- e) for mismatch type N (see Table 5) up to 0,10 mm, the minimum acceptable radius at base of mismatch (i.e: the minimum cutter tool radius) could be 0,5 mm.

NOTES

- 1) For mismatch type M according Table 5, the surface could be blended smooth by sanding since the requirements of a, b, c and d are met. See item 7.1.
- 2) Mismatch type B could adopt mismatch type N acceptance criteria, since it is positioned at distances greater than three times the corner internal radius or the bottom internal radius, whichever is greater.

TABLE 5 : GENERAL TOLERANCES FOR MISMATCH



MISMATCH TYPE	DESCRIPTION	TOLERANCE (MAXIMUM VALUE) mm
A	MISMATCH ON INTERSECTION BETWEEN TWO SURFACES AS SHOWN ON FIGURE	0,8 (3)
B	MISMATCH ON AN INDEPENDENT INTERNAL SURFACE.	0,20 (3)
C	MISMATCH ON INTERSECTION (ADJACENT ACUTE ANGLE AND CONCORDANCE RADIUS) BETWEEN TWO INCLINED SURFACES.	0,8 FOR $T \geq 8$ $T/10$ FOR $T < 8$ (2) (3)

continued

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TABLE 5 : GENERAL TOLERANCES FOR MISMATCHES

continued

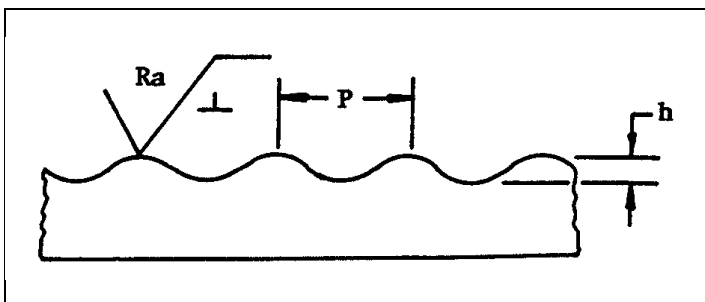
MISMATCH TYPE	DESCRIPTION	TOLERANCE (MAXIMUM VALUE) mm
D	INCLINED MISMATCH PRODUCED BY END OF MILLING CUTTER ON INTERSECTION (ADJACENT TO ACUTE ANGLE) BETWEEN TWO INCLINED SURFACES.	AS ALLOWED ON DRAWING
E	MISMATCH ON INTERSECTION (ADJACENT TO CONCORDANCE RADIUS) BETWEEN TWO PERPENDICULAR SURFACES.	0,50 FOR $T \geq 5$ $T/10$ FOR $T < 5$ (2) (3)
F	MISMATCH ON INTERSECTION (DEPRESSION SHAPE) BETWEEN TWO PERPENDICULAR SURFACES.	0,20 FOR $T \geq 2$ $T/10$ FOR $T < 2$ (2)
G	MISMATCH ON INTERSECTION (DEPRESSION SHAPE ADJACENT TO OBTUSE ANGLE) BETWEEN TWO INCLINED SURFACES.	0,20 FOR $T \geq 2$ $T/10$ FOR $T < 2$ (2)
H	F MISMATCH WITH A MINIMUM WIDTH OF 8 MILIMETERS.	0,20 FOR $T \geq 2$ $T/10$ FOR $T < 2$ (2)
J	MISMATCH ON EXTERNAL CORNER RADIUS.	0,50
K	MISMATCH ON CONCORDANCE RADIUS OF EXTERNAL ROUND SURFACES.	0,20 (3)
L	MISMATCH ON CONCORDANCE RADIUS OF INTERNAL ROUND SURFACES	0,50 (3)
M	MISMATCH ON BEARING FASTENING SURFACES, FAYING OR SEATING SURFACES	0,05 (3)
N	MISMATCH ON AN INDEPENDENT EXTERNAL SURFACE.	0,20 (4)

- (1)** Radius obtained by the bottom radius of milling cutter, since greater than 2 mm.
(2) T = detail thickness.
(3) The types A, B, C, E, K, L and M mismatch radius value shall be the same as the bottom internal radius value specified on drawing.
(4) Type N mismatch base radius value shall be according sub item 5.4.4 (d) or 5.4.4 (e).

5.4.5 WAVING

Waving tolerances shall be according to Table 6.

TABLE 6 : GENERAL TOLERANCES FOR WAVING

	SURFACE ROUGHNESS SPECIFIED ON DRAWING Ra (μm)	h (MAX) mm	P mm
	$Ra \leq 0,8$	0,02	0,10
	$0,8 < Ra \leq 1,2$	0,02	0,14
	$1,2 < Ra \leq 3,2$	0,06	0,38

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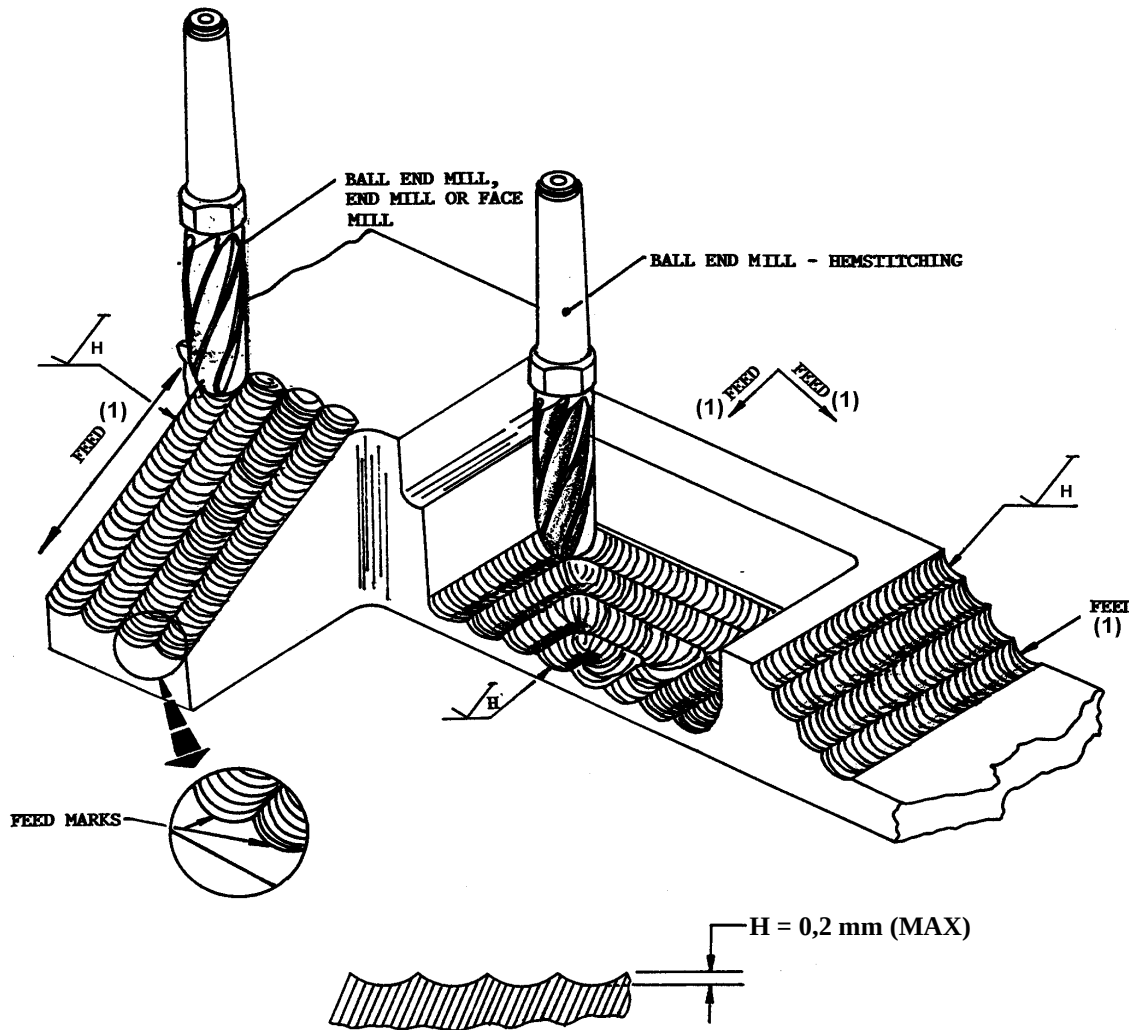
5.4.6 HEMSTITCHING

Hemstitching tolerances, if permitted on drawing (see item 7.1), shall be according to Figure 2.

NOTES

1) Whenever the symbol $\sqrt{H}$ is not indicated on drawing, a maximum value of 0,05 mm for hemstitching is acceptable and the surface could be blended smooth by sanding.

2) The hemstitching radius value produced by the end of milling in an inclined surface, shall be the same as the bottom internal radius value specified on drawing.



HEMSTITCHING HEIGHT (H)

(1) The R_a shown by the drawing shall be measured in the direction of feed. Additional finishing such as grinding or sanding is not necessary (and usually not desirable) if the hemstitching surface is within dimensional limits of the drawing.

FIGURE 2 : GENERAL TOLERANCE OF TYPICAL SURFACE TEXTURE - HEMSTITCHING

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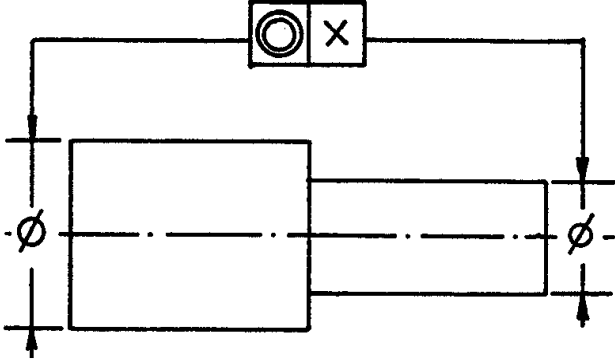
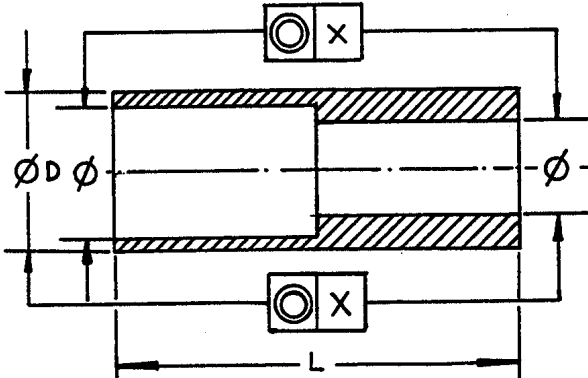
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5.4.7 CONCENTRICITY DEVIATIONS

Tolerance for concentricity deviations shall be according to Table 7.

TABLE 7 : GENERAL TOLERANCES FOR CONCENTRICITY DEVIATIONS

DEVIATION TYPE	TOLERANCE mm
- DEVIATION BETWEEN EXTERNAL DIAMETERS 	$X = 0,05$ (FOR ANY DIMENSION)
- DEVIATION BETWEEN INTERNAL DIAMETERS - DEVIATION BETWEEN EXTERNAL AND INTERNAL DIAMETERS 	$X = 0,050$ FOR $\frac{L}{\phi D} > 2$ $X = 0,025$ FOR $\frac{L}{\phi D} \leq 2$

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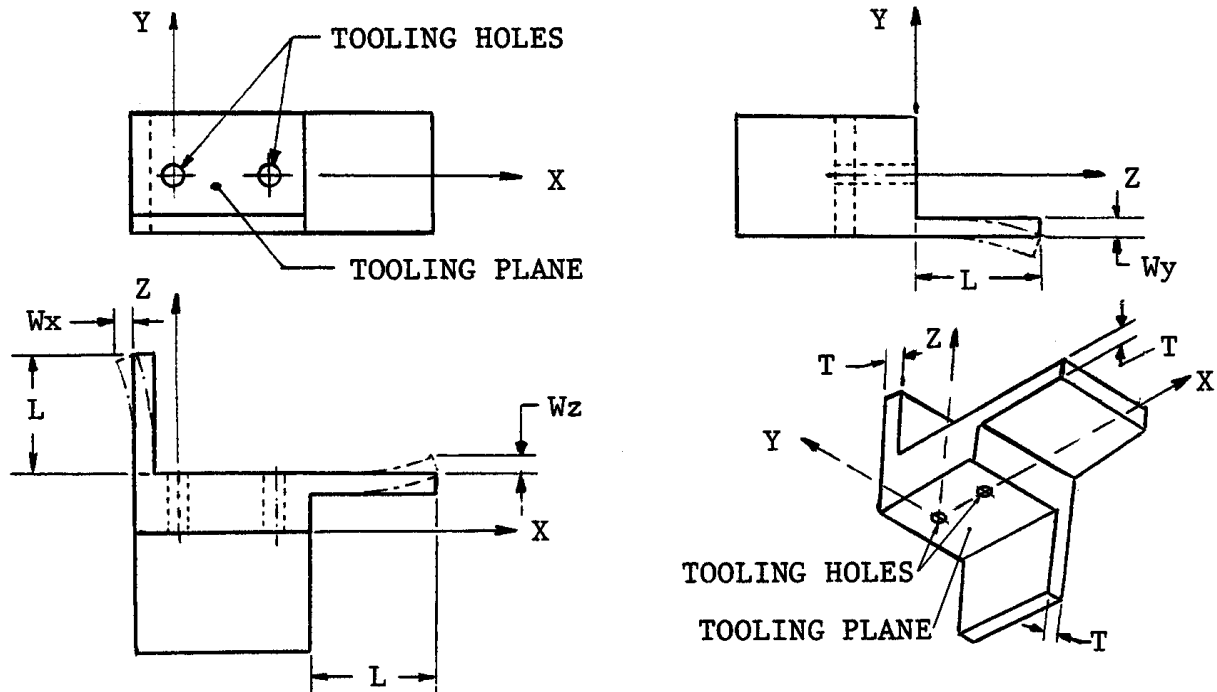
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5.4.8 WARPING

Tolerance for warping shall be according to the following criteria (see Figure 3) :



Acceptable: is possible to locate all surface points into the 250 mm partial length in a 0,25 mm gap

Not acceptable: is not possible to locate all surface points into the 250 mm partial length in a 0,25 mm gap

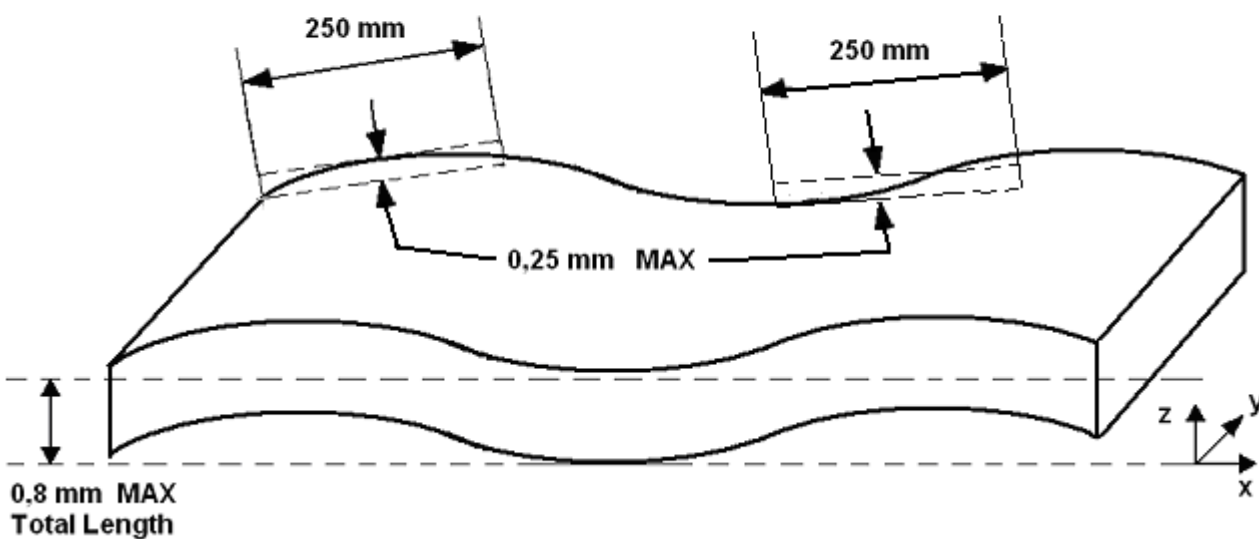


FIGURE 3 : GENERAL TOLERANCE FOR WARPING

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- $W_x (\text{max}) = W_y (\text{max}) = W_z (\text{max}) = 0,8 \text{ mm}$ of warping for the total part length in any direction.
- For any partial length of 250 mm the maximum warping tolerance shall be 0,25 mm. Short partial lengths could be used to facilitate measuring operation, since the ratio partial length/warping tolerance shall be kept the same.
- For parts with length $L < 250 \text{ mm}$, the maximum warping tolerance shall be $L/1000$.

USE OF WEIGHTS

If the warping on first measurement is larger than the tolerance value, the measurement shall be repeated using weights as follows:

- a) 1 kgf for thickness $(T) \leq 1 \text{ mm}$;
- b) 2 kgf for thickness $(T) > 1 \text{ mm}$.

The weights shall be applied on the warping detail on two perpendicular arranged directions, at 250 mm intervals (one to the other) or proportionally for smaller intervals. Example: half of the specified weight at 125 mm intervals.

There are two tolerances for weight application:

- a) 0,05 mm maximum for places just under the weight locations; and
- b) the warping general tolerance used for measurement, without weight application, for the other places.

NOTES

1) Before warping measurement, the part shall be supported by a device that represents its assembling on the aircraft (it shall be used the part datums, such as tooling holes, tooling planes or other important details for the part assembling) and the warping value shall be the gap between a point on the real part surface and the place that it should be according the drawing.

2) See item 7.1.5.

5.4.9 FLATNESS

The total flat surface of the part shall lie between two planes 0,8 mm apart and additionally, to any 250 mm diameter circle the part surface shall lie between two parallel planes 0,25 mm apart according shown in Figure 4.

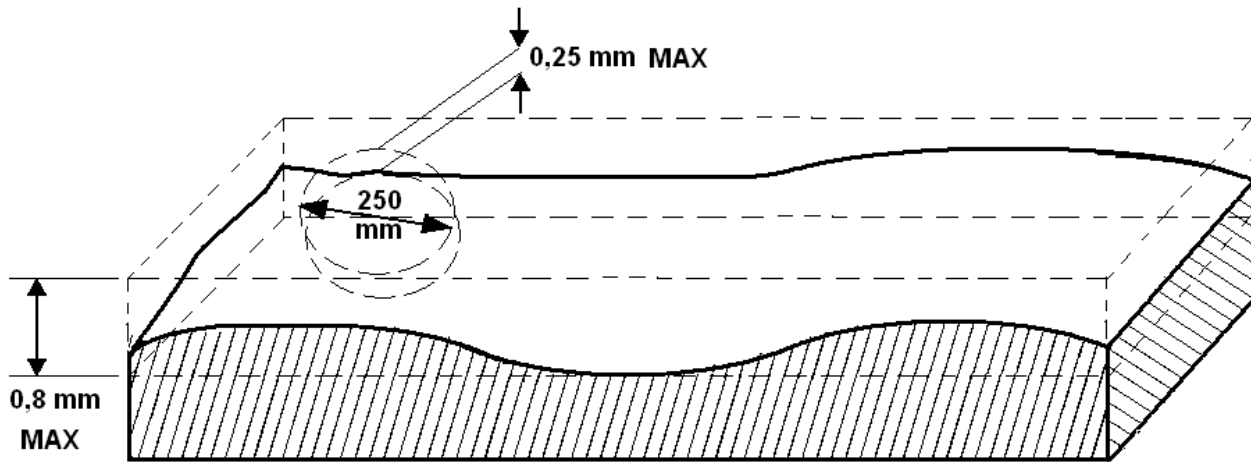


FIGURE 4 : TOLERANCE FOR FLATNESS

5.5 GRAIN FLOW DIRECTION

Grain flow direction tolerance (general or specific) is applicable to machined parts made of laminated, forged or extruded material.

Consider the following cases:

a) drawing does not indicate grain direction, but calls out NI 383 (refer to item 7.1). In this case, grain direction is optional and may be established according to manufacturing convenience;

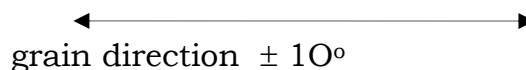
b) drawing indicates grain direction but does not indicate specific angular tolerance. In this case, general angular tolerance to be considered shall be $\pm 15^\circ$;

EXAMPLE:



c) drawing indicates grain direction and a specific angular tolerance. In this case, indicated tolerance is mandatory.

EXAMPLE:



5.6 SANDING OPERATIONS

Whenever necessary to obtain the desired countour or to correct any geometric deviation, a sanding operation with aluminum oxide sandpaper could be used. Sandpaper grit equal or finer than those specified in Table 8 shall be used according to the operation purpose and a finishing operation (FO) is mandatory over the surface where a processing operation (PO) is realized for coarse roughing.

TABLE 8 : SANDPAPER GRIT FOR SANDING OPERATION

GRIT (1)	ALUMINIUM ALLOYS	STEELS	TITANIUM ALLOYS
100	-	PO	PO / FO
150	PO / FO	FO	-

(1) according CAMI (Coated Abrasives Manufacturer's Institute) standard.

NOTES

- The finishing operation shall be done to remove all scratches due the previous sanding of the processing operation.
- An adequate sandpaper grit sequence from coarse to thinness, shall be used to obtain the specified requirements.
- After the rework by sanding, the part shall comply with all requirements specified on drawing.
- For aluminum alloys only: whenever necessary an aluminum oxide disc with equivalent grit (Roloc Scotch-Brite color Red: 150-180 and Blue: 220-320) may be used as substitute for sandpaper.

6. QUALITY ASSURANCE PROVISIONS

6.1 The individual responsible for performing machined parts shall be trained and shall guarantee the fulfillment of the requirements of this Standard.

6.2 Embraer Quality Assurance shall monitor the process of machined parts by any auditing process so as to guarantee the fulfillment of item 6.1 and promote preventive actions when necessary.

7. GENERAL NOTES

7.1 DRAWING CALLOUT

7.1.1 This Standard shall be called out on the production drawings through the "NI 641 - General tolerances for machined metallic details according to NE 03-071.

7.1.2 Indicate, if applicable the "NI 383 - Grain Direction not indicated is optional".

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7.1.3 Indicate the symbol $\sqrt{H}$ on drawing where hemstitching greater than 0,05 mm is permitted. See Figure 2.

7.1.4 If mismatch is critical on a specific area (fastener area, for example), it shall be indicated: "NI 774 : Mismatch tolerance on this area according NE 03-071, Table 5, mismatch type M".

7.1.5 If the standard values of general tolerance (0,8 mm or L/1000 mm) and/or weight (1 kgf or 2 kgf) are not adequate to check warping (see item 5.4.8), the following Item Notes (NI) shall be called out on drawing:

"NI 805 : Maximum allowable warping $< * >$ mm and warping control procedures according to NE 03-071".

* - indicate:

-a "specific value" for warping tolerance or

- "NA" (Not Applicable) for parts that do not need warping control.

If there are more than one "specific value" for warping, indicate the first one using NI 805 and the others using Free Notes (NL) with the same writing of NI 805.

"NI 804 : Maximum weight application $< * >$ kgf at the indicated points and warping control procedures according to NE 03-071".

* indicate the maximum weight value.

If there are more than one "weight value", indicate the first one using NI 804 and the others using Free Note (NL) with the same writing of NI 804.

7.1.6 If a sheet part is mechanical milled instead chemical milled, the following note shall be used:

"NI 1531 : Specific thickness tolerance for machined sheet parts according to NE 03-071".

The use of this NI implies in the accomplishment of all other requirements of this Standard, and the parts shall be submitted to penetrant inspection to guarantee no surface cracks or heat damage. The following note shall be used:

"NI 624 : Liquid penetrant inspection according to NE 07-001"

8. BIBLIOGRAPHIC REFERENCES

Report 000EBD005 - "Análise do Processo de Usinagem Mecânica em Chapas de Alumínio".

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MACHINED METALLIC DETAILS - GENERAL TOLERANCES

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